LAB REPORT – TASK 1

Student Information:

SHAH KHALID

Electrical Eng (cammunication)

Registration No: BF25NWELE0672

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Section B

Title:

Study of Different Data Types, Their Memory Size, Location, and Values in C++

Objective:

The objective of this experiment is to understand different built-in data types in C++, declare variables of each type, and determine their memory size, stored values, and memory locations.

Theory:

In C++, variables are used to store data in memory. Each variable has:

- A **data type** (which defines the type of data it can store)
- A **value** (the data stored in it)
- A **memory location** (address where it is stored)

Different data types require different amounts of memory. Some common data types are:

- **int:** Used to store whole numbers (e.g., 10, -5)
- **float:** Used for decimal numbers with single precision
- **double:** Used for decimal numbers with higher precision
- **char:** Used to store a single character

- **bool:** Used to store logical values (true or false)

The **`sizeof()` operator** is used to determine the memory size (in bytes) of a variable.
The **address-of operator (`&`)** is used to access the memory location of a variable.

Understanding memory size and location is important in programming, especially in fields like embedded systems and optimization.

Program (with comments):

```
#include <iostream>   // Library for input and output operations
using namespace std;  // Allows use of standard names without std::

int main()
{
    // Declaration and initialization of variables
    int a = 10;          // Integer variable storing whole number
    float b = 5.5;       // Float variable storing decimal number
    double c = 10.12345; // Double variable for higher precision decimal
    char d = 'A';        // Character variable storing single character
    bool e = true;       // Boolean variable storing true/false

    // Display integer information
    cout << "Integer value: " << a << endl;                 // Prints value
of a
    cout << "Size of int: " << sizeof(a) << " bytes" << endl; // Prints
memory size
    cout << "Address of int: " << &a << endl << endl;       // Prints memory
address

    // Display float information
    cout << "Float value: " << b << endl;
    cout << "Size of float: " << sizeof(b) << " bytes" << endl;
    cout << "Address of float: " << &b << endl << endl;

    // Display double information
    cout << "Double value: " << c << endl;
    cout << "Size of double: " << sizeof(c) << " bytes" << endl;
    cout << "Address of double: " << &c << endl << endl;

    // Display character information
    cout << "Char value: " << d << endl;
    cout << "Size of char: " << sizeof(d) << " bytes" << endl;
    cout << "Address of char: " << &d << endl << endl;

    // Display boolean information
    cout << "Bool value: " << e << endl;
    cout << "Size of bool: " << sizeof(e) << " bytes" << endl;
    cout << "Address of bool: " << &e << endl;
```

```
    return 0;  // Program ends successfully
}
```

Explanation:

In this program, five variables of different data types are declared and initialized. The program uses `cout` to display their values. The `sizeof()` operator calculates how many bytes each variable occupies in memory. The `&` operator prints the memory address, which is different for each variable.

Output:

(Integer value: 10

Size of int: 4 bytes

Address of int: 0x6ffe0c

Float value: 5.5

Size of float: 4 bytes

Address of float: 0x6ffe08

Double value: 10.1235

Size of double: 8 bytes

Address of double: 0x6ffe00

Char value: A

Size of char: 1 bytes

Address of char: Aσaí╓4?$@

Bool value: 1

Size of bool: 1 bytes

Address of bool: 0x6ffdfe)

Conclusion:

This experiment demonstrates that different data types occupy different amounts of memory and are stored at unique memory locations. Understanding this helps in writing efficient and optimized programs.